ASSIGNMENT-1
DBMS

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B.tech CSE (CORE)-A

SECTION-A

Ques-1 What is Database Management System? Why is it required?

Ans-1 A Database Management System is software that is used to store, manage, organize, and retrieve data efficiently in a structured manner.
It acts as an interface between the user and the database.

DBMS is required

1. To reduce data redundancy
2. To maintain data consistency
3. To provide data security
4. To allow easy access and retrieval
5. To support backup and recovery
6. To handle multiple users

DBMS helps in managing data in a secure and organized way.

Ques-2 Why is DBMS better than the traditional file System?

Ans-2 The traditional file system stores data in separate files, which creates many problems.

Problems in traditional file System

1. Data redundancy
2. Data inconsistency
3. Lack of security
4. No proper concurrency control
5. No backup and recovery system.

Advantages of DBMS over file System

1. Reduces Redundancy
2. Avoids inconsistency
3. Provides security
4. Support concurrency
5. Provides backup and recovery

DBMS is more efficient and reliable than the traditional file system.

Ques-3 What is Data Abstraction? Explain its levels.

Ans-3 Data Abstraction means hiding the internal details of the database and showing only necessary information to users.

There are three levels of data abstraction:

1. Physical level
   Describes how data is stored in memory.
2. Logical level
   Describes what data is stored and relationship between them
3. View level
   Describes what the user sees.

Data abstraction improves security and simplicity.

Ques-4 What is the three-Schema Architecture? Why is it important?

Ans-4 The three-Schema Architecture is a database design that divides the database into three levels:

1. Internal Schema - Physical storage details.
2. Conceptual Schema - Logical structure of the database.
3. External Schema - User view level

Importance:

1. Provides data independence.
2. Separates user view from physical storage.
3. Makes database structured and organized.
4. Allows easy modification.

Ques-5 What is Data Independence? Why is it needed?

Ans-5 Data Independence is the ability to change the schema at one level without affecting the other levels.

Types of data Independence

1. Physical Data Independence
   changes in physical level do not affect logical level.
2. Logical Data Independence
   changes in logical level do not affect view level.

why it is needed.
Provides flexibility
Makes system easy to modify.
Simplifies maintenance

Q-6 What is an Entity-Relationship [ER] Model?

Ans-6 An Entity-relationship model is a conceptual model used to represent real-world objects in a database.

It shows:

1. Entities: (obj)
2. Attributes: (properties of entities)
3. Relationships: (connection between entities)

Ques-7 What are Integrity Constraints? Why are they important in DBMS?

Ans-7 Integrity constraints are rules that maintain data accuracy and consistency in a database.

Examples:

1. Primary key
2. Foreign key
3. Unique
4. Not Null
5. Check

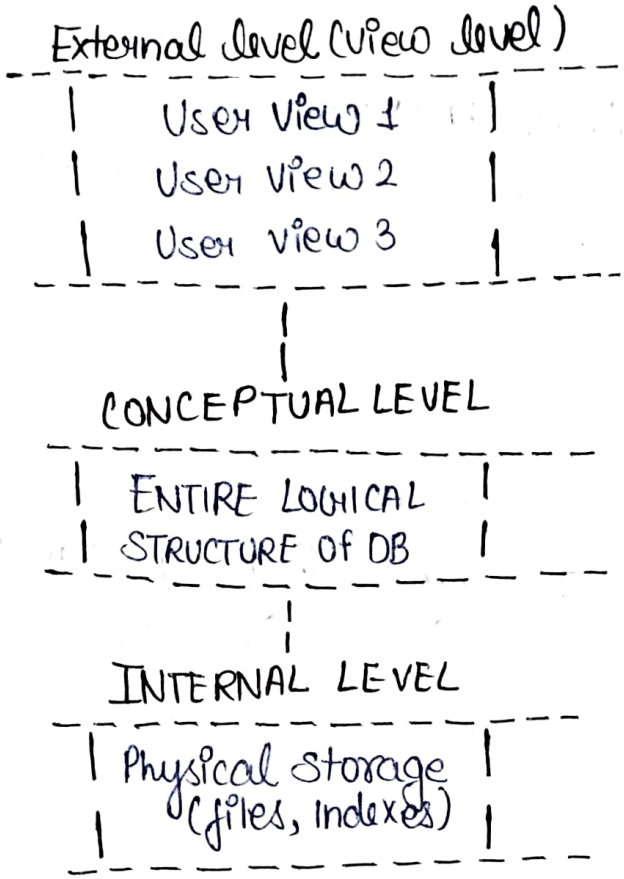
Importance

1. Prevents invalid data entry.
2. Maintains data consistency.
3. Ensures database reliability.

SECTION - B

Ques-1 Draw the three - schema architecture and explain each level.

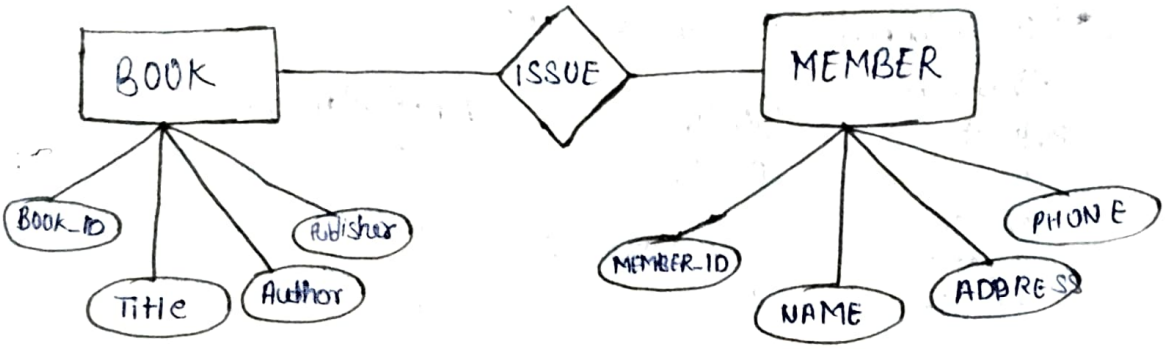
Ans-1



1. External level
   1. Highest level of abstraction.
   2. Shows only required data to users.
   3. Different users can have different views.
   4. Provides Security.
2. Conceptual level
   1. Describes the complete logical structure of the database.
   2. Define entities, attributes, relationships.
   3. Independent of physical storage.
3. Internal level
   1. Lowest level of abstraction.
   2. Describes how data is stored in memory.
   3. Includes files organization, indexing, storage details

Importance:
1. Provides data independence.
2. Separates user view from storage.
3. Makes database structured and flexible.

Ques-2 Draw an ER diagram for a library Management System.



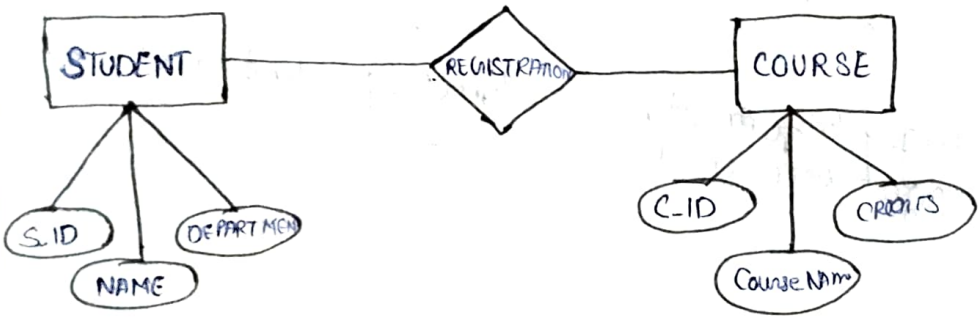
ENTITIES:
1. BOOK
2. MEMBER
3. ISSUE

ATTRIBUTES:

BOOK	MEMBER
BOOK_ID	MEMBER_ID
Title	Name
Author	Address
Publisher	Phone

RELATIONSHIP:
- Member borrows book
- One member can borrow many books
- One book can be borrowed by many members over time.

Ques-3 Draw an ER diagram for a Student-Course Registration System.



ENTITIES:
STUDENT
COURSE
Registration

ATTRIBUTES

STUDENT	COURSE
Student_ID	COURSE_ID
Name	course
Department	Credits

Relationship:
- A student can register for many course.
- A course can have many students.
- Cardinality: Many-to-many.

SECTION - C

Ques-1 A hospital wants to store patient, doctor and appointment data. Identify entities, attributes and relationships.

ENTITIES:
- Patient
- Doctor
- Appointment

Attributes:

Patient
- Patient_ID
- NAME
- GENDER
- PHONE

Doctor
- Doctor_ID
- NAME
- SPECIALIZATION
- PHONE

Appointment
- Appointment_ID
- Date
- Time
- Patient_ID
- Doctor_ID

Relationship
- A patient books Appointment.
- A Doctor attends Appointment.
- One patient can have many appointment
- One doctor can attend many appointment.

Constraints:
- Patient_ID must exist in Patient_table.
- Doctor_ID must exist in Doctor_table.
- Appointment date cannot be null.
- Primary keys must be unique.

Question-2 An online shopping platform stores customer, product and order data. Suggest keys and constraints.

Entities:
1. Customer
2. Product
3. Order

Keys.

Customer
- Customer_ID (PK)
- Email (Unique constraint)

Product
- Product_ID
- Product_Name
- Price

Order

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date

Integrity Constraints

- Customer_ID in Order must exist in Customer table
- Price must be greater than 0
- Email must be unique
- Primary keys must not be null
- Quantity must be positive.

Ques-3 A university stores student records. Explain how data abstractio helps different users.

Ans-3 Data abstraction hides internal details and shows only required information to users.

There are three levels.

1. View level
   1. Student see only their marks and personal details.
   2. Teacher see marks of their subjects.
   3. Admin sees full student information

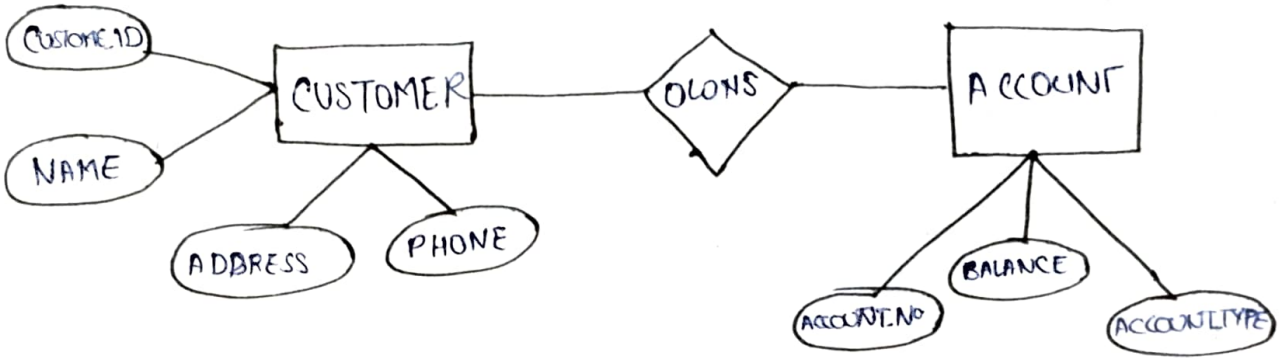
2. Logical level
   1. Describes complete structure of database.
   2. Contains Student table, Course table, Result table
   3. Defines relationships between them.

3. Physical level
   1. Describes how data is stored in memory.
   2. Includes file structure and indexing
   3. Hidden from users.

SECTION- E

SECTION- D

Ques-1 Draw a simple ER diagram for any real-life system and explain one key design decision.



ENTITIES

1. CUSTOMER
2. ACCOUNT

RELATIONSHIP

- A customer owns an Account.
- One customer can have many accounts.
- One account belongs to one customer.

One key design

Decision: Use customer-ID as Primary key instead of Name.

Justification

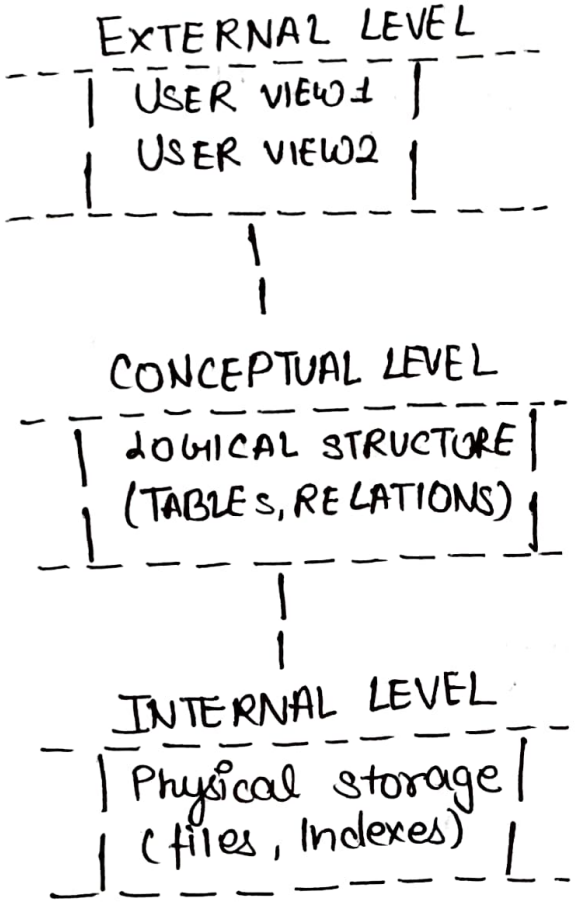
- Names can be same for different customers.
- customer-ID is unique.
- Avoids confusion and duplication.
- Ensures data accuracy.

Ques-2 Draw the three-Schema architecture and mention one benefit.

Ans-2 Benefit: Data Independence

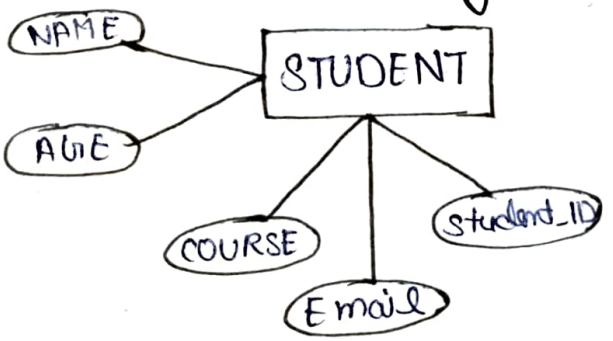
changes at the physical level do not affect logical or view level.

This makes the system flexible and easy to maintain.



Ques-3 Draw an entity with attributes and Identify its primary key.

Ans-3 Ex→ Student Entity



Entity – Student

Attributes – Student_ID, Name, Age, course, Email

Primary key – Student_ID

Student_ID is a primary key because

1. It uniquely identifies
2. No two student can have same Student_ID
3. It can't be NULL.

SECTION- E

Ques-1 Write two Assumptions you made while Drawing an ER Diagram

Ans-1 Assumption 1:

I assume that each entity has a unique Primary key to uniquely identify each record.

Ex - Student_ID

Assumption 2:

I assume proper cardinality between entities.

Ex → One student can register for many courses, but each registration belongs to one student.

Ques-2 Mention One limitation of ER Modeling

Ans-2 One major limitation of ER modeling is that

- It only represents the conceptual structure of the database.
- It does not show implementation details like indexing, storage methods, or performance factors.
- Complex business rules are sometimes difficult to represent clearly.

Ques-3 What part of DBMS Unit-1 Did you most difficult and why?

Ans-3 The most difficult part of DBMS unit-1 was understanding Data Independence and three-schema Architure because:

It involves multiple abstraction levels

It requires understanding the difference b/w physical, logical and view levels.

The relationship between schemas can be confusing initially.